

Script-Controlled ACL – Restrict Record Access Based on Field Value

Enterprise Data Security & Dynamic Access Control Framework

Project Team Overview

TEAM ID & SIZE

SWTID-2026-2854

Total Members: 5

TEAM LEADER

MOHAMED ISTHI A

mohamedisthi19@gmail.com

Role	Member Name	Email Address
Team Member 1	Amudhan	amudhansrinivasan02@gmail.com
Team Member 2	DHARUN KUMARP	dd3156684@gmail.com
Team Member 3	PREMKUMAR.N	nrpremkumar67@gmail.com
Team Member 4	PRIZEGOD V	prizegod2616@gmail.com

1. Brainstorming & Ideation Phase

1 Conceptualization & Problem Definition

Standard role-based access control (RBAC) often falls short in enterprise platforms where record visibility must adapt dynamically depending on row-level field contents (e.g., status, confidentiality tier, or departmental assignment). The team brainstormed script-controlled Access Control Lists (ACLs) using JavaScript conditions to evaluate record fields securely on read, write, create, and delete actions.

2. Requirement Analysis Phase

2 Functional & Technical Specifications

Identified core platform constraints, JavaScript security sandboxing rules, and performance metrics. Requirements established that script evaluation must execute efficiently without creating latency overhead, supporting granular context checking against user roles and active record field values.

3. Project Design Phase

3 Architecture & Logic Flow Design

Designed the architectural workflow linking user session data, record object models, and script evaluation scripts (``answer = checkAccess();``). The module design cleanly separates field-level conditional rules from global table configurations.

4. Project Planning Phase

4 Milestones & Resource Allocation

Formulated a strict timeline dividing tasks among all 5 team members. Mohamed Isthi A directed overall coordination and script validation, while Amudhan, Dharun Kumar P, Premkumar N, and Prizegod V managed module design, backend script writing, unit testing, and documentation respectively.

5. Project Development Phase

5 Implementation & Script Coding

Developed server-side JavaScript scripts inside ACL rules to interrogate field values dynamically. Implemented robust error handling to prevent script timeouts or silent access blocks, ensuring reliable evaluation across diverse data sets.

6. Project Testing Phase

6 Validation & Security Verification

Executed comprehensive test cases simulating different user roles attempting to access records with varied sensitive field values. Verified that unauthorized access attempts are correctly blocked while permitted users retain seamless access.

7. Project Documentation Phase

7 Technical Manuals & Guides

Compiled comprehensive configuration manuals, code repositories, administrator guidelines, and troubleshooting flowcharts. Ensured complete transparency and maintainability for future deployments.

8. Project Demonstration Phase

8 Presentation & Live Showcase

Successfully demonstrated the working Script-Controlled ACL solution to reviewers. Showcased real-time record restriction behavior driven dynamically by changing field attributes under various security profiles.

Conclusion

The **Script-Controlled ACL – Restrict Record Access Based on Field Value** project successfully achieves a secure, highly customizable data access barrier. Through structured collaboration across all eight project phases, Team SWTID-2026-2854 delivered a robust enterprise-grade security solution.